Student Name: Student ID:

ELEC 473 – Cryptography and Network Security
Quiz 3

Wednesday, Nov. 15, 2023, 2:30-3:00 PM

INSTRUCTIONS—Read Carefully

• Write your name and ID legibly at the top of the papers.

• Calculators are allowed to use.

• Show all your work. Partial credit will be given for substantial progress towards the solution. No credit will be given for answers with no substantial process.

Part I Multiple Choices - please mark the best answer (subtotal: 10 marks).

1. Suppose a prime $p \equiv 3 \bmod 4$ and $a \in Z_p$. Which of the following is equivalent to a square root of a mod p?

A. $a^{p-1} \bmod p$

B. $a^{(p+1)/4} \bmod p$

C. $a^{(p+1)/2} \bmod p$

D. $a^{(p-1)/2} \bmod p$

2. How many bit-output is generated by SHA-1?

A. 512

B. 256

C. 160

D. 128

3. Which one of the following statements about hash function is incorrect?

A. If a hash function is strong collision resistant, it must be second pre-image resistant.

B. If a hash function is weak collision resistant, it must be second pre-image resistant.

C. If an adversary can find a collision of the hash function, he must be able to solve the second pre-image problem.

D. If an adversary can solve the preimage problem of the hash function, he must be able to find a collision.

4. Let C(K, M) denote a message authentication code function, produced for the message M and a shared key K. Let E(K, M) denote encryption of a message M with a key K, and let || denote the concatenation. If Alice sends to Bob the following information: $E(K_2, M \| C(K_1, M))$ where K_1 and K_2 are shared secret keys, it is
A. just message authentication
B. message authentication and confidentiality where authentication is tied to the plaintext
C. message authentication and confidentiality where authentication is tied to the ciphertext
D. message authentication and confidentiality where authentication is tied both to the plaintext and to the ciphertext

5. A receiver must be able to check a message to prevent the adversary from modifying it during transmission, it is known as
A. Message integrity
B. Message non-repudiation
C. Message confidentiality
D. Message sending

6. The elliptic curve group is defined by $y^2 = x^3 - 17x + 16$ over real numbers, what is P + Q, if P = (0, -4) and Q = (1, 0)?
A. (15, -56)
B. (-23, -43)
C. (69, 26)
D. (12, -86)

7. Which of the following schemes can provide authenticity and integrity?
A. SHA-3
B. AES
C. MD5
D. ECDSA

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8. Which of the following is an advantage of public-key cryptography over symmetric-key cryptography?
A. Public-key cryptography provides more security services.
B. Public-key cryptography does not rely on conjectured hardness of certain computational problems.
C. Public-key cryptography has higher throughput.
D. Public-key cryptography has shorter key size.

9. Cryptographic hash function takes an arbitrary block of data and returns
A. fixed size bit string
B. variable size bit string
C. both fixed size bit string and variable size bit string
D. none of the mentioned

10. Which of the following statements about the security of digital signatures are false?
A. Without the private signing key, it is computationally infeasible for an attacker to construct a signature, such that the verification algorithm can “accept” the signature.
B. The attacker cannot produce a valid signature based on chosen message attacks.
C. The attacker cannot produce a valid signature based on known-message attacks.
D. Secure digital signatures should prevent the existential forgery in which an attacker can create a valid signature on a given message, selected by the signer.

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Part II (subtotal: 20 marks).

Q1. (3+3+2 marks)

a) What are the three cryptographic mechanisms that are used to protect the integrity of messages?

b) What are the key differences between these three mechanisms?

c) Comparing the properties of these three mechanisms, which one can provide best protection on emails, why?

a) Message Authentication Code (MAC) 1 mark

Hash function 1 mark

Digital signature 1 mark

b. MAC is in symmetric key model, should pre-share a secret key. 1 mark

Hash function does not have any key 1 mark

Digital signature is in asymmetric key (public key) model, do not need to pre-share the key. (A signer uses the private key to sign a message and the verifier uses the public key to verify the signature). 1 mark

c. Digital signature 1 mark

because do not need to pre-share the key and can achieve authenticity, integrity, and non-repudiation. 1 mark

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Q2. (4 marks) Describe the Diffie-Hellman key agreement protocol to show how Alice and Bob negotiate a shared secret key.

Alice 1 mark

choose random **a** in {1,…,q}

1 mark **Bob**

choose random **b** in {1,…,q}

$A = g^a \longrightarrow$

$\longleftarrow B = g^b$

$\mathbf{B^a} = (g^b)^a = \mathbf{k_{AB} = g^{ab}} = (g^a)^b = \mathbf{A^b}$

1 mark 1 mark.

Q3. (8 marks) Let E denote the elliptic curve $y^2 \equiv x^3+x+6$, defined over Z_{11}. It can be shown that $\#E=13$. Considering the Elliptic Curve ElGamal cryptosystem that is implemented in E. P=(2, 7) is the generator of E(Z_{11}) and a=5 is the private key of Alice.

a. Compute the public key of Alice.
b. Suppose Bob needs to send a message m=(3, 5) to Alice. Bob chooses a random value k=7 for encryption. Compute the corresponding ciphertext of m.
c. Show the decryption of the ciphertext in b from Bob.
d. If the message m=(10, 2) and the random value k=3, what is the ciphertext?

(Hint: P= (2,7), 2P= (5,2), 3P= (8,3), 4P= (10,2), 5P= (3,6), 6P= (7,9), 7P= (7,2), 8P= (3,5), 9P=(10,9), 10P= (8,8), 11P= (5,9), 12P=(2,4))

a. PK=aP=5(2, 7)=(3, 6) 1 mark for 5(2, 7) 1 mark for (3, 6)
b. U=kP=7(2, 7)=(7, 2) 1 mark for (7, 2)
V=m+kPK=(3, 5)+7(3, 6)=(3, 5)+(10, 9)=(10, 2) 1 mark for (10, 2)
c. m=V-aU=(10, 2)-5(7, 2)=(10, 2)-(10, 9)=2(10, 2)=(3, 5) 1 mark for (10, 2)-5(7, 2) 1 mark for (3, 5)
d. U=kP=3(2, 7)=(8, 3) 1 mark for (8, 3)
V=m+kPK=(10, 2)+3(3, 6)=(10, 2)+(5, 2)=(7, 9) 1 mark for (7, 9)